

Project-Helix — Entity-Relationship (ER) Specification

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1. 9-Table 3NF Normalized Relational Database Schema

The architecture strictly models the 9 tables presented in Slide 32 of Project_Helix_DBMS_25B11CS893.pptx:

- **USERS:** user_id (PK), name, username (UK), email (UK), password_hash, role, created_at
- **STARTUPS:** startup_id (PK), name, description, industry, founded_date, status, created_by (FK)
- **TEAM_MEMBERS:** team_id (PK), startup_id (FK), user_id (FK), role_in_team, joined_date
(Associative Entity)
- **JOBS:** job_id (PK), startup_id (FK), title, description, status, posted_date
- **PROJECTS:** project_id (PK), startup_id (FK), name, description, start_date, end_date, status
- **TASKS:** task_id (PK), project_id (FK), title, description, assigned_to (FK), status, due_date, priority
- **FUNDING_ROUNDS:** round_id (PK), startup_id (FK), round_type, target_amount, status, round_date
- **INVESTORS:** investor_id (PK), name, email (UK), phone, firm_name
- **INVESTMENTS:** investment_id (PK), investor_id (FK), round_id (FK), amount, investment_date
(Associative Entity)

2. Entity Classification & Normalization

- **7 Strong Entities:** USERS, STARTUPS, JOBS, PROJECTS, TASKS, FUNDING_ROUNDS, INVESTORS.
- **2 Associative Entities:** TEAM_MEMBERS (resolves M:N Users-Startups) & INVESTMENTS (resolves M:N Investors-Rounds).
- **Normalization:** 100% compliant with Third Normal Form (3NF) with 0 transitive dependencies.